

CYBER SECURITY INTERNSHIP REPORT

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Domain: Ethical Hacking

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Projects and Tasks (Beginner Level)

Assessment 4: Network Vulnerability Scanning using Nmap

1. Introduction

The objective of this task is to perform network reconnaissance using Nmap in order to identify open ports, running services, and potential vulnerabilities in a target system.

Network scanning is a crucial phase in penetration testing, as it helps in understanding the attack surface of a system. In this task, a vulnerable machine (Metasploitable) was scanned using different Nmap techniques to gather detailed information about its network services.

2. About Nmap

Nmap (Network Mapper) is an open-source tool used for network discovery and security auditing.

Key Capabilities:

- Host discovery
 - Port scanning
 - Service and version detection
 - Operating system identification
 - Script-based vulnerability scanning
-

3. Lab Environment

- **Attacker Machine:** Kali Linux
 - **Target Machine:** Metasploitable
 - **Target IP Address:** 192.168.56.101
 - **Tools Used:** Nmap, Terminal
 - **Network Type:** Virtual lab (Host-only Adapter)
-

4. Methodology

The scanning process was performed in multiple stages to gather detailed information about the target system.

4.1 Installation of Nmap

```
sudo apt update  
sudo apt install nmap -y
```

4.2 Target Identification

The target machine was identified using network configuration tools, and its IP address was confirmed as:

```
ifconfig
```

```
192.168.56.101
```

4.3 Basic TCP Scan

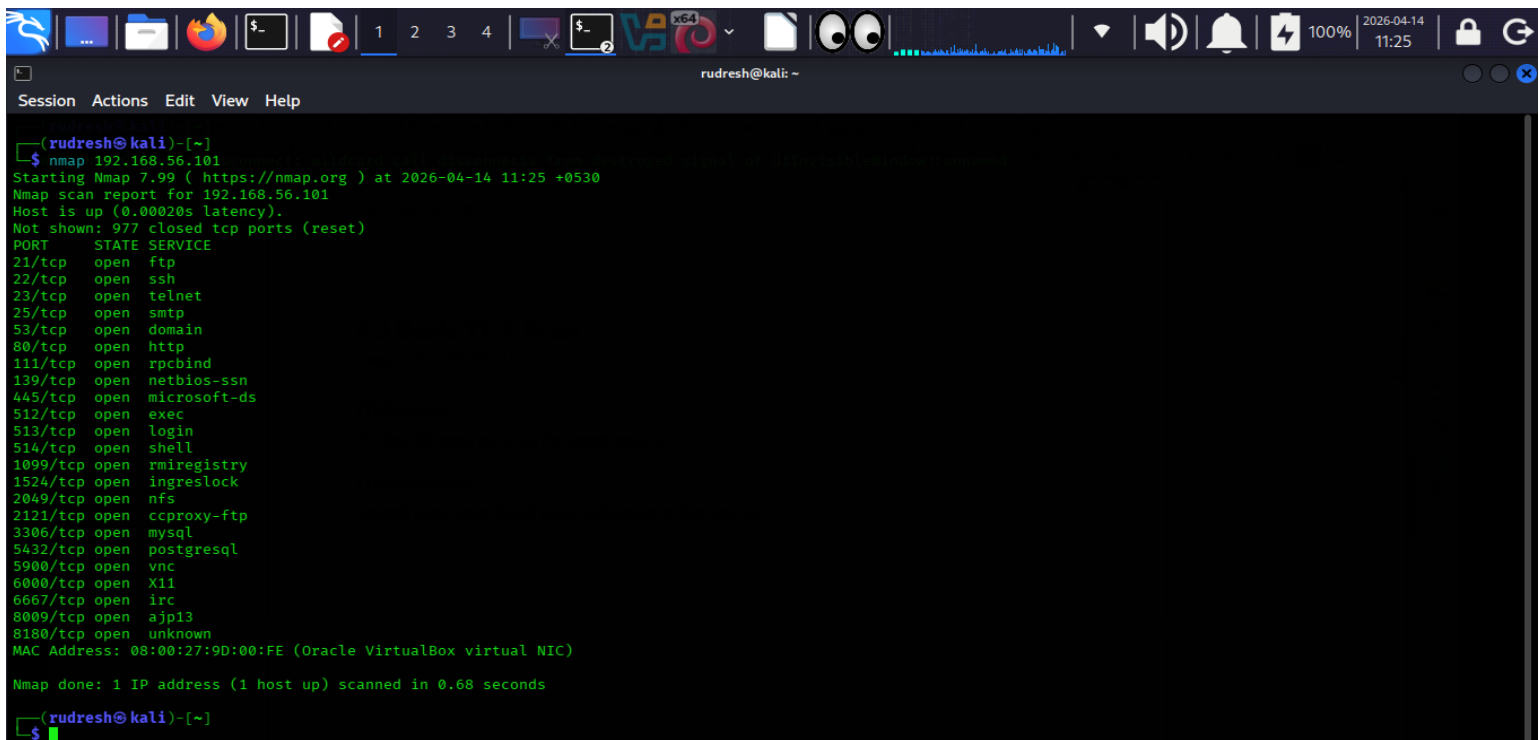
```
nmap 192.168.56.101
```

Objective:

To identify open ports on the target system.

Observation:

Several ports were found open, indicating active service



```
(rudresh@kali)-[~]
$ nmap 192.168.56.101
Starting Nmap 7.99 ( https://nmap.org ) at 2026-04-14 11:25 +0530
Nmap scan report for 192.168.56.101
Host is up (0.00020s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
23/tcp    open  telnet
25/tcp    open  smtp
53/tcp    open  domain
80/tcp    open  http
111/tcp   open  rpcbind
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
512/tcp   open  exec
513/tcp   open  login
514/tcp   open  shell
1099/tcp  open  rmiregistry
1524/tcp  open  ingreslock
2049/tcp  open  nfs
2121/tcp  open  ccproxy-ftp
3306/tcp  open  mysql
5432/tcp  open  postgresql
5900/tcp  open  vnc
6000/tcp  open  X11
6667/tcp  open  irc
8009/tcp  open  ajp13
8180/tcp  open  unknown
MAC Address: 08:00:27:9D:00:FE (Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 0.68 seconds

(rudresh@kali)-[~]
$
```

4.4 Service Version Detection

`nmap -sV 192.168.56.101`

Objective:

To determine the versions of services running on open ports.

Observation:

Outdated versions of services such as FTP and Apache were identified.

```
Session Actions Edit View Help

(rudresh@kali) ~
$ nmap -sV 192.168.56.101
Starting Nmap 7.99 ( https://nmap.org ) at 2026-04-14 11:27 +0530
Nmap scan report for 192.168.56.101
Host is up (0.00025s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE      VERSION
21/tcp    open  ftp          vsftpd 2.3.4
22/tcp    open  ssh          OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
23/tcp    open  telnet       Linux telnetd
25/tcp    open  smtp         Postfix smtpd
53/tcp    open  domain       ISC BIND 9.4.2
80/tcp    open  http         Apache httpd 2.2.8 ((Ubuntu) DAV/2)
111/tcp   open  rpcbind      2 (RPC #100000)
139/tcp   open  netbios-ssn  Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
445/tcp   open  netbios-ssn  Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
512/tcp   open  exec         netkit-rsh rexecd
513/tcp   open  login        NetKit rshd
514/tcp   open  shell        GNU Classpath grmiregistry
1099/tcp  open  java-rmi     Metasploitable root shell
1524/tcp  open  bindshell    2-4 (RPC #100003)
2049/tcp  open  nfs          ProFTPD 1.3.1
2121/tcp  open  ftp          MySQL 5.0.51a-3ubuntu5
3306/tcp  open  mysql        PostgreSQL DB 8.3.0 - 8.3.7
5432/tcp  open  postgresql   VNC (protocol 3.3)
5900/tcp  open  vnc          (access denied)
6000/tcp  open  X11          UnrealIRCd
6667/tcp  open  irc          Apache Jserv (Protocol v1.3)
8009/tcp  open  ajp13        Apache Tomcat/Coyote JSP engine 1.1
8180/tcp  open  http         (detection output)
MAC Address: 08:00:27:90:00:FE (Oracle VirtualBox virtual NIC)
Service Info: Hosts: metasploitable.localdomain, irc.Metasploitable.LAN; OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/.
Nmap done: 1 IP address (1 host up) scanned in 12.20 seconds
```

4.5 Operating System Detection

```
nmap -O 192.168.56.101
```

Objective:

To identify the operating system of the target machine.

Observation:

The target system was detected as a Linux-based OS, consistent with Metasploitable.

```
rudresh@kali: ~  
$ nmap -O 192.168.56.101  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-04-14 11:28 +0530  
Nmap scan report for 192.168.56.101  
Host is up (0.00025s latency).  
Not shown: 977 closed tcp ports (reset)  
PORT      STATE SERVICE  
21/tcp    open  ftp  
22/tcp    open  ssh  
23/tcp    open  telnet  
25/tcp    open  smtp  
53/tcp    open  domain  
80/tcp    open  http  
111/tcp   open  rpcbind  
139/tcp   open  netbios-ssn  
445/tcp   open  microsoft-ds  
512/tcp   open  exec  
513/tcp   open  login  
514/tcp   open  shell  
1099/tcp  open  rmiregistry  
1524/tcp  open  ingreslock  
2049/tcp  open  nfs  
2121/tcp  open  ccproxy-ftp  
3306/tcp  open  mysql  
3432/tcp  open  postgresql  
5900/tcp  open  vnc  
5900/tcp  open  X11  
5667/tcp  open  irc  
8009/tcp  open  ajp13  
8180/tcp  open  unknown  
MAC Address: 08:00:27:9D:00:FE (Oracle VirtualBox virtual NIC)  
Device type: general purpose  
Running: Linux 2.6.X  
OS CPE: cpe:/o:linux:linux_kernel:2.6  
OS details: Linux 2.6.9 - 2.6.33  
Network Distance: 1 hop  
OS detection performed. Please report any incorrect results at https://nmap.org/submit/ .  
Nmap done: 1 IP address (1 host up) scanned in 1.95 seconds
```

4.5 Operating System Detection

`nmap -O 192.168.56.101`

Objective:
To identify the operating system of the target machine.

Observation:
The target system was detected as a Linux-based OS, consistent with Metasploitable.

4.6 Aggressive Scan

4.6 Aggressive Scan

`nmap -A 192.168.56.101`

Objective:

To perform an advanced scan combining:

- OS detection
- Version detection
- Script scanning
- Traceroute

Observation:

Detailed information about services, versions, and potential vulnerabilities was obtained.

```
Session Actions Edit View Help

(rudresh@kali) ~ -
$ nmap -A 192.168.56.101
Starting Nmap 7.99 ( https://nmap.org ) at 2026-04-14 11:29 +0530
Nmap scan report for 192.168.56.101
Host is up (0.00027s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE      VERSION
21/tcp    open  ftp          vsftpd 2.3.4
| ftp-syst:
|   STAT:
|   FTP server status:
|     Connected to 192.168.56.1
|     Logged in as ftp
|     TYPE: ASCII
|     No session bandwidth limit
|     Session timeout in seconds is 300
|     Control connection is plain text
|     Data connections will be plain text
|     vsFTPd 2.3.4 - secure, fast, stable
|_ End of status
|_ ftp-anon: Anonymous FTP login allowed (FTP code 230)
22/tcp    open  ssh          OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
|_ ssh-hostkey:
|   1024 60:0f:cf:e1:c0:5f:6a:74:d6:90:24:fa:c4:d5:6c:cd (DSA)
|   2048 56:56:24:0f:21:1d:de:a7:2b:ae:61:b1:24:3d:e8:f3 (RSA)
23/tcp    open  telnet       Linux telnetd
25/tcp    open  smtp         Postfix smtpd
|_ sslv2:
|   SSLv2 supported
|   ciphers:
|     SSL2_DES_64_CBC_WITH_MD5
|     SSL2_RC2_128_CBC_WITH_MD5
|     SSL2_RC4_128_EXPORT40_WITH_MD5
|     SSL2_RC2_128_CBC_EXPORT40_WITH_MD5
|     SSL2_RC4_128_WITH_MD5
|     SSL2_DES_192_EDE3_CBC_WITH_MD5
|_ ssl-date: 2026-04-14T05:59:45+00:00; -2s from scanner time.
|_ ssl-cert: Subject: commonName=ubuntu804-base.localdomain/organizationName=OCOSA/stateOrProvinceName=There is no such thing outside US/countryName=XX
|_ Not valid before: 2010-03-17T14:07:45

9. Conclusion
This task provided practical exposure to network vulnerability scanning using Nmap. By performing different types of scans, it was possible to identify open ports, detect running services, and analyze potential vulnerabilities in the target system. This knowledge is essential for both offensive and defensive cybersecurity practices.

10. Final Outcome
```

```
Session Actions Edit View Help

(rudresh@kali) ~ -
80/tcp    open  http         Apache httpd 2.2.8 ((Ubuntu) DAV/2)
|_ http-server-header: Apache/2.2.8 (Ubuntu) DAV/2
|_ http-title: Metasploitable2 - Linux
111/tcp    open  rpcbind      2 (RPC #100000)
|_ rpcinfo:
|   program version  port/proto  service
|   100000  2           111/tcp     rpcbind
|   100000  2           111/udp     rpcbind
|   100003  2,3,4       2049/tcp    nfs
|   100003  2,3,4       2049/udp    nfs
|   100005  1,2,3       53551/tcp   mountd
|   100005  1,2,3       53674/udp   mountd
|   100021  1,3,4       34578/udp   nlockmgr
|   100021  1,3,4       39486/tcp   nlockmgr
|   100024  1           42723/tcp   status
|   100024  1           54865/udp   status
139/tcp    open  netbios-ssn  Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
445/tcp    open  netbios-ssn  Samba smbd 3.0.20-Debian (workgroup: WORKGROUP)
512/tcp    open  exec         netkit-rsh rexecd
513/tcp    open  login        Netkit rshd
514/tcp    open  shell        Netkit rshd
1099/tcp   open  java-rmi     GNU Classpath grmiregistry
1524/tcp   open  bindshell    Metasploitable root shell
2049/tcp   open  nfs          2-4 (RPC #100003)
2121/tcp   open  ftp          ProFTPD 1.3.1
3306/tcp   open  mysql        MySQL 5.0.51a-3ubuntu5
|_ mysql-info:
|   Protocol: 10
|   Version: 5.0.51a-3ubuntu5
|   Thread ID: 9
|   Capabilities flags: 43564
|   Some Capabilities: Support41Auth, ConnectWithDatabase, SwitchToSSLAfterHandshake, SupportsTransactions, LongColumnFlag, SupportsCompression, Speaks41ProtocolNew
|   Status: Autocommit
|   Salt: *(#91737Z?k!)\rurTo
5432/tcp   open  postgresql   PostgreSQL DB 8.3.0 - 8.3.7
|_ ssl-date: 2026-04-14T05:59:45+00:00; -2s from scanner time.
|_ ssl-cert: Subject: commonName=ubuntu804-base.localdomain/organizationName=OCOSA/stateOrProvinceName=There is no such thing outside US/countryName=XX
|_ Not valid before: 2010-03-17T14:07:45
|_ Not valid after: 2010-04-16T14:07:45
```

5. Results and Analysis

The scan results revealed multiple open ports and services running on the target system.

5.1 Open Ports and Services

Port	Service	Description	Security Risk
21	FTP	File transfer service	May allow anonymous access or backdoor
22	SSH	Secure remote login	Vulnerable to brute-force attacks
23	Telnet	Remote access	Unencrypted, highly insecure
80	HTTP	Web server (Apache)	Vulnerable to web attacks
139/445	SMB	File sharing	Exploitable via SMB vulnerabilities
3306	MySQL	Database service	Risk of unauthorized access

5.2 Service Version Analysis

The version detection scan revealed outdated services such as:

- vsftpd (FTP server)
- Apache web server

These outdated versions are known to contain vulnerabilities and can be exploited by attackers.

5.3 OS Detection Analysis

The OS detection scan indicated that the target system is running a Linux-based operating system, which matches the expected configuration of Metasploitable.

5.4 Aggressive Scan Findings

The aggressive scan provided:

- Detailed service information
- Script-based vulnerability results
- Network path information

This helped in identifying weak points in the system.

6. Vulnerability Assessment

Based on the scan results:

- **FTP (vsftpd):**
 - Known for backdoor vulnerabilities
- **Telnet:**
 - Transmits data in plain text

- **Apache (HTTP):**
 - Outdated version vulnerable to exploits
 - **SMB:**
 - Susceptible to remote code execution attacks
-

7. Security Implications

The presence of multiple open ports and outdated services significantly increases the attack surface of the system.

Key risks include:

- Unauthorized access
 - Data interception
 - Remote exploitation
 - Credential theft
-

8. Skills Acquired

- Network reconnaissance using Nmap
 - Port and service enumeration
 - Vulnerability identification
 - Understanding attack surfaces
 - Interpreting scan results
-

9. Conclusion

This task provided practical exposure to network vulnerability scanning using Nmap. By performing different types of scans, it was possible to identify open ports, detect running services, and analyze potential vulnerabilities in the target system.

This knowledge is essential for both offensive and defensive cybersecurity practices.

10. Final Outcome

- Successfully scanned the target system
- Identified open ports and services

- Detected OS and service versions
 - Analyzed vulnerabilities and risks
-